

Himanshu Ambalal Patil

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SUMMARY

Software Engineer with 3+ years of experience building scalable backend applications using Java, Spring Boot, MongoDB, and SQL. Improved API and query performance by up to 10× and reduced manual effort by up to 90% through automation and system optimization. Experienced in REST APIs, microservices, CI/CD pipelines, and enterprise software development.

EDUCATION

University at Buffalo, SUNY | Master of Science: Computer Science and Engineering **Aug 2024 - Dec 2025**

Course work: Data Intensive Computing, Introduction to Machine Learning, Deep Learning, Intro to Pattern Recognition, DMQL

SGGSIE&T, India | Bachelor of Technology: Computer Science and Engineering **Aug 2017 - Jul 2021**

Course work: OOP with Java, Data Structures, ADBMS, Software Engineering, Big Data Analytics, Probability and Statistics.

TECHNICAL SKILLS

- **Programming & Development:** Java, Python, C, SQL, JavaScript, HTML, CSS, Spring Boot, Spring MVC, Hibernate.
- **Backend Technologies:** Core Java, Spring Boot, Spring MVC, Spring Data JPA, Hibernate, RESTful APIs, Microservices, Apache Kafka, OOP, Collections, Multithreading, SOLID Principles
- **Databases:** MongoDB, PostgreSQL, MySQL, JDBC (SQL & NoSQL design, query optimization)
- **DevOps & Tools:** AWS, CI/CD, Jenkins, Docker, Git, GitHub, Github Copilot, Jira, VS Code, MongoDB Compass, Eclipse, IntelliJ.
- **Additional Skills:** Data Structures & Algorithms, Machine Learning, Deep Learning, NLP

WORK EXPERIENCE

Kaiburr | Software Engineer **Apr 2022 – Jul 2024**

- Developed Java backend microservices using Spring Boot, JPA, and MongoDB to ingest metadata from tools like Jira, ADO, Bitbucket, and Jenkins, processing data for over 100+ projects across enterprise clients while applying SOLID principles for clean and maintainable code.
- Designed and implemented backend processing services using Java, Groovy, and Python to transform, aggregate, and process enterprise-scale data from multiple third-party integrations while enabling reusable business logic across applications.
- Improved backend application performance by up to 10× by redesigning MongoDB aggregation pipelines, implementing targeted indexing strategies, and optimizing query execution paths, reducing query execution time by 90% across high-volume enterprise workloads.
- Authored and standardized complex MongoDB aggregation pipelines, powering multi-filter, reusable dashboards through a single extensible query framework used across products.
- Reduced dashboard development effort by 90%, introducing parameterized queries that dynamically adapted to new filters, KPIs, and use cases without additional backend changes.
- Built automation services using Python to generate and validate database queries, reducing manual configuration effort by 50% and minimizing production errors.
- Identified and resolved backend and data-layer bottlenecks affecting latency and scalability, ensuring low-latency analytics and reliable dashboard rendering under increasing data volume.
- Recognized within 6 months as the primary owner for analytics performance and MongoDB optimization, mentoring new joiners, assigning and reviewing tasks, establishing team-wide best practices, and delivering technical training on scalable query design and dashboard architecture.

Cognizant Technology Solutions | Programmer Analyst Trainee **Aug 2021 - Mar 2022**

- Implemented and maintained RESTful APIs and backend services using Java, Spring Boot, Spring MVC, and Hibernate for enterprise-grade applications, ensuring scalable and modular backend design.
- Developed full-stack web features using HTML, CSS, and JavaScript, enhancing UI consistency and improving user experience.
- Modeled and tuned MySQL schemas, indexes, and queries to enable scalable and reliable backend data access.
- Applied advanced Core Java concepts including OOP, Collections, Multithreading, and Exception Handling to solve 2+ real-world case study problems.

PROJECT

Distributed E-Commerce Platform | Java, Spring Boot, Microservices, API Gateway ([Link](#)) **Jan 2026 – Mar 2026**

- Designed and developed a distributed e-commerce backend system using Spring Boot microservices architecture with 3 independent services (Product, Order, Inventory), enabling modular and scalable system design.
- Implemented Spring Cloud Gateway as a centralized API Gateway to route and manage all client requests across microservices, standardizing entry-point request handling and improving system structure.
- Built 15+ RESTful APIs using a layered Controller–Service–Repository architecture, where Order Service interacts with Inventory Service for stock validation and order processing, ensuring loosely coupled and maintainable backend design.

KnowYourDoc | Hackathon Project ([Link](#)) **Feb 2025 - Feb 2025**

- Architected an end-to-end legal document analysis platform using FastAPI, PyPDF2, Gemini API, and a responsive HTML/CSS/JavaScript frontend to support PDF uploads, text extraction, and interactive insight visualization.
- Reduced manual document review effort by ~60% by implementing robust JSON parsing and validation logic to surface structured outputs such as summaries, complexity ratings, red flags, extracted figures, and loopholes.

F1 Race Winner Predictor | Machine Learning ([Link](#)) **Aug 2024 - Dec 2024**

- Constructed a machine learning pipeline to predict Formula 1 race winners using historical Grand Prix data, including driver performance, team metrics, grid positions, and pit stop strategies.
- Trained and evaluated five models (Logistic Regression, KNN, Random Forest, XGBoost, Neural Network), achieving best recall of 0.78 with XGBoost through GridSearchCV tuning and cross-validation to address class imbalance.